

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices and Circuits Laboratory Report

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Course Code: EE2301

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1 Objective

1. To plot the forward and reverse bias V-I characteristics **PN junction (rectifier) diode** and to determine its cut-in voltage, static and dynamic forward resistance, and reverse saturation current
2. To plot the forward and reverse bias V-I characteristics of a **Zener diode** and to determine its **Zener breakdown voltage** (V_z) and **Zener resistance** (R_z)
3. To design, assemble, and test positive, negative, and biased **Clipper circuits** , and positive and negative **Clamper circuits**, and to record their input-output waveforms.

2 Part A - PN Junction Diode: Forward Bias

2.1 Components Used

S.No.	Component / Equipment	Specification	Quantity
1	PN Junction Diode	1N4007	1
2	DC Power Supply	0–30 V	1
3	Digital Multimeter	–	2
4	Resistors	1 k Ω ,	1
5	Breadboard	–	1
16	Connecting Wires and Probes	–	As required

2.2 Circuit

The PN junction diode was connected in series with a 1 k Ω current-limiting resistor and a milliammeter. The voltage across the diode was measured using a voltmeter connected directly across the diode.

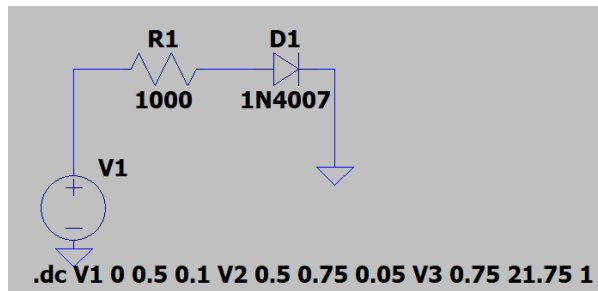


Figure 1: Circuit for forward-bias characteristic of PN junction diode

2.3 Observation Table

2.3.1 PN Junction Diode — Forward Bias

Table 1: Forward Bias Characteristics of PN Junction Diode

S.No.	V_{in} (V)	V_f (V)	I_f (mA)
1	0.00	0.000	0.000
2	0.10	0.100	0.000
3	0.20	0.201	0.000
4	0.30	0.301	0.000
5	0.40	0.399	0.002
6	0.50	0.478	0.024
7	0.55	0.499	0.053
8	0.60	0.520	0.082
9	0.65	0.535	0.120
10	0.70	0.549	0.158
11	0.75	0.552	0.199
12	2.85	0.651	2.1925
13	2.86	0.652	2.2011
14	4.90	0.679	4.2070
15	6.92	0.693	6.2045
16	8.94	0.704	8.203
17	10.95	0.712	10.1987
18	12.97	0.719	12.203
19	14.98	0.729	14.200
20	17.00	0.729	16.204
21	19.01	0.733	18.200
22	21.02	0.736	20.198

2.4 Simulations

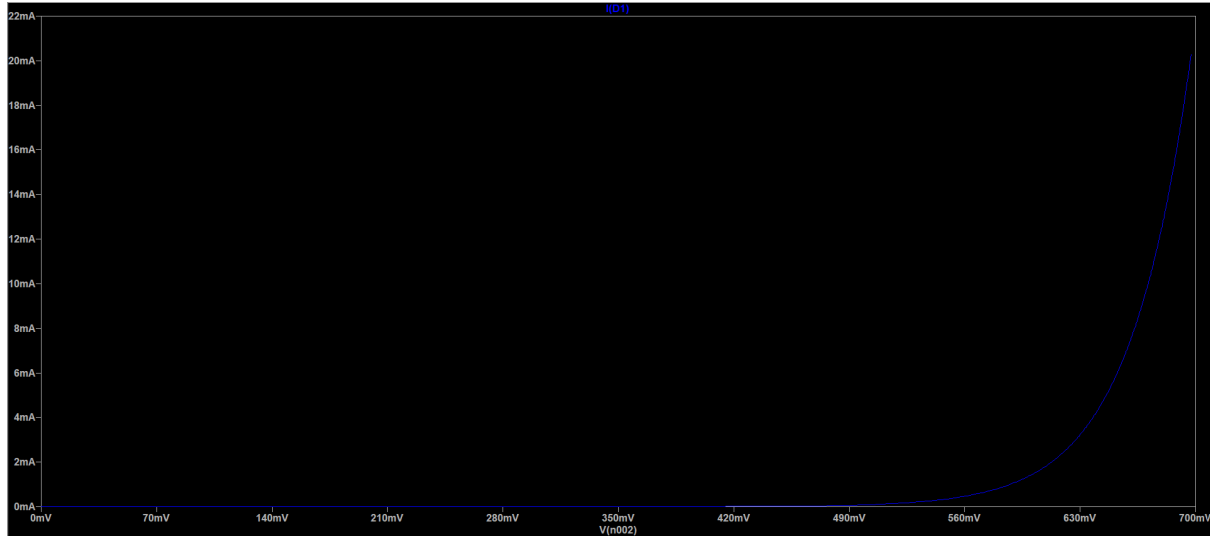


Figure 2: I_f vs V_f

2.5 Calculations and Analysis

2.5.1 Forward-Bias Characteristics of PN Junction Diode

The forward characteristic of the PN junction diode is obtained by plotting the forward current I_f on the Y-axis against the forward voltage V_f on the X-axis.

2.5.2 Cut-in Voltage

The cut-in voltage (or knee voltage) V_γ is the minimum forward voltage beyond which the forward current starts increasing rapidly.

From the observed forward-bias characteristics, the diode current remains very small up to approximately 0.55 V. A significant increase in current is observed beyond approximately 0.65 V. For example,

$$V_F = 0.552 \text{ V}, \quad I_F = 0.199 \text{ mA}$$

whereas

$$V_F = 0.651 \text{ V}, \quad I_F = 2.1925 \text{ mA}.$$

Therefore, from the knee region of the observed characteristic, the cut-in voltage is approximately

$$V_\gamma \approx 0.65 \text{ V}$$

This value is consistent with the typical cut-in voltage of a silicon PN junction diode.

2.5.3 Static Forward Resistance

The static forward resistance of the diode at a particular operating point is given by

$$R_F = \frac{V_F}{I_F}.$$

From the observation table, consider the operating point

$$V_F = 0.704 \text{ V}$$

and

$$I_F = 8.203 \text{ mA}.$$

Therefore,

$$R_F = \frac{0.704}{8.203 \times 10^{-3}}$$

$$R_F = 85.82 \Omega.$$

Hence, the static forward resistance at this operating point is

$$\boxed{R_F \approx 85.82 \Omega}$$

2.5.4 Dynamic Forward Resistance

The dynamic forward resistance is obtained from the slope of the forward characteristic at a particular operating point.

It is given by

$$r_f = \frac{\Delta V_F}{\Delta I_F}.$$

Using two nearby points from the observation table,

$$(V_{F1}, I_{F1}) = (0.704 \text{ V}, 8.203 \text{ mA})$$

and

$$(V_{F2}, I_{F2}) = (0.712 \text{ V}, 10.1987 \text{ mA}),$$

we obtain

$$r_f = \frac{0.712 - 0.704}{(10.1987 - 8.203) \times 10^{-3}}$$

$$r_f = \frac{0.008}{1.9957 \times 10^{-3}}$$

$$r_f \approx 4.01 \, \Omega.$$

Therefore,

$$\boxed{r_f \approx 4.01 \, \Omega}$$

at the selected operating point.

2.5.5 Diode Equation

The forward current of a PN junction diode is approximately described by the Shockley diode equation

$$I_F = I_0 \left[\exp \left(\frac{V_F}{\eta V_T} \right) - 1 \right],$$

where

I_0 = reverse saturation current,

$$V_T = \frac{kT}{q},$$

and η is the ideality factor.

At room temperature,

$$V_T \approx 25.9 \, \text{mV}.$$

The equation shows that the forward current increases exponentially with the applied forward voltage. This behavior is clearly observed in the experimental data, where the current increases from less than 0.2 mA near 0.55 V to more than 20 mA near 0.74 V.

2.5.6 Analysis

From the observed forward characteristic, the diode current remains very small at low forward voltages. The current begins to increase significantly in the knee region around

$$\boxed{V_\gamma \approx 0.65 \, \text{V}}.$$

Beyond this region, a small increase in forward voltage produces a large increase in forward current, which demonstrates the nonlinear nature of the PN junction diode.

At the operating point

$$V_F = 0.704 \text{ V}, \quad I_F = 8.203 \text{ mA},$$

the static forward resistance is

$$R_F \approx 85.82 \Omega,$$

while the dynamic forward resistance calculated using two nearby operating points is

$$r_f \approx 4.01 \Omega.$$

The dynamic resistance is considerably smaller than the static resistance because it represents the local slope of the nonlinear forward characteristic.

Thus, the experimental observations confirm the expected forward-bias behavior of a silicon PN junction diode.

2.5.7 Graphs from observation

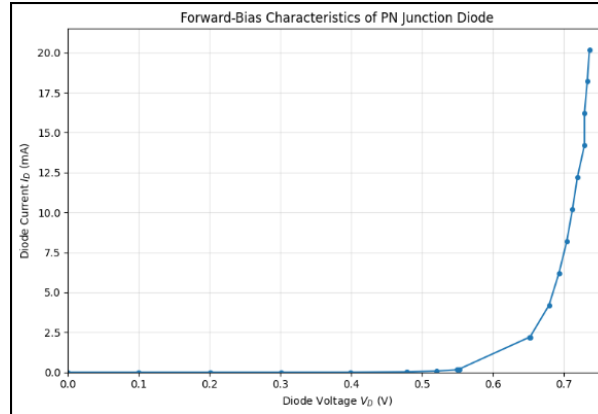


Figure 3: both x and y axis are linear

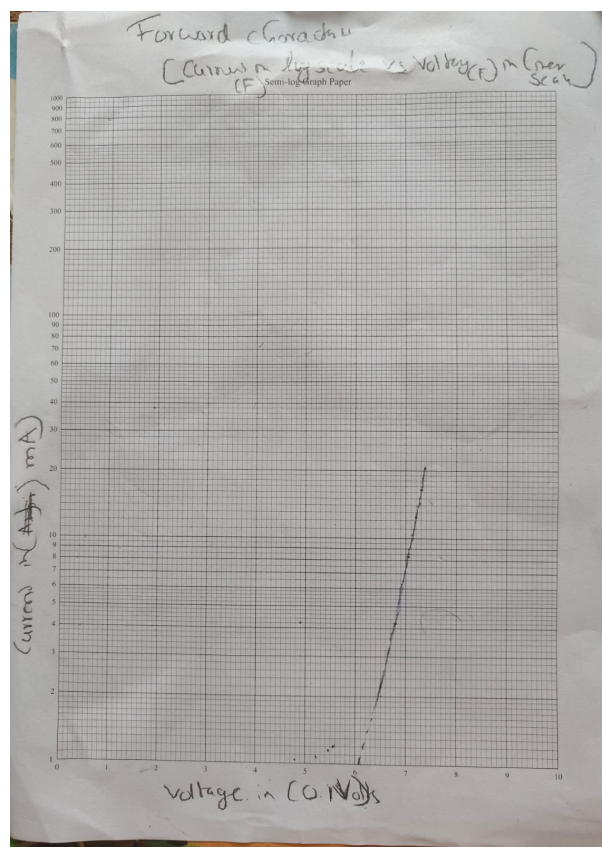


Figure 4: current in log scale and voltage in linear scale

3 Part B — PN Junction Diode: Reverse Bias

3.1 Components Used

S.No.	Component / Equipment	Specification	Quantity
1	PN Junction Diode	1N4007	1
2	DC Power Supply	0–30 V	1
3	Digital Multimeter	—	2
4	Resistors	1 k Ω ,	1
5	Breadboard	—	1
16	Connecting Wires and Probes	—	As required

3.2 Circuit

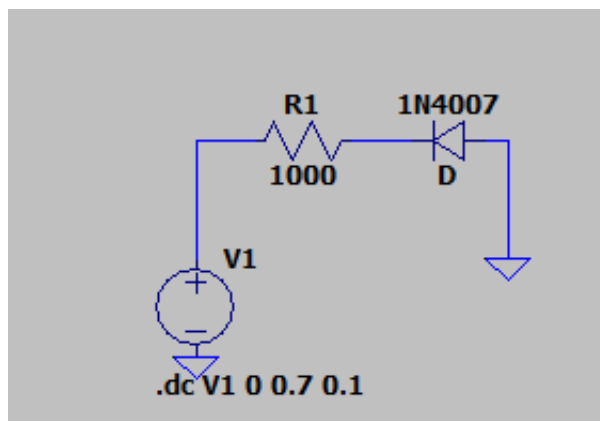


Figure 5

3.3 Observation Table

Table 2: Reverse Bias Characteristics of Diode

Input Voltage V_{DC}	Voltage V_Z	Current I_Z
5	5.03	0.0004
10	10.05	0.0009
15	15.07	0.0017
20	20.10	0.0020
25	25.13	0.0024
30	30.15	0.0028

3.4 Simulations



Figure 6: Current across the diode vs voltage across the diode

Note = there is blue line passing through the 0 therefore denoting current almost zero

3.5 Calculations for Reverse Bias

For the reverse-biased PN junction diode, the diode current is calculated using Kirchhoff's Voltage Law:

$$V_{DC} = I_D R + V_D$$

Therefore,

$$I_D = \frac{V_{DC} - V_D}{R}$$

where R is the series resistance.

3.5.1 Calculation for $V_{DC} = 0.7\text{ V}$

From the experimental data,

$$V_{DC} = 0.7\text{ V}$$

and

$$V_D = 0.703\text{ V}$$

Hence,

$$I_D = \frac{V_{DC} - V_D}{R}$$

$$I_D = \frac{0.700 - 0.703}{R}$$

$$I_D = \frac{-0.003}{R}\text{ A}$$

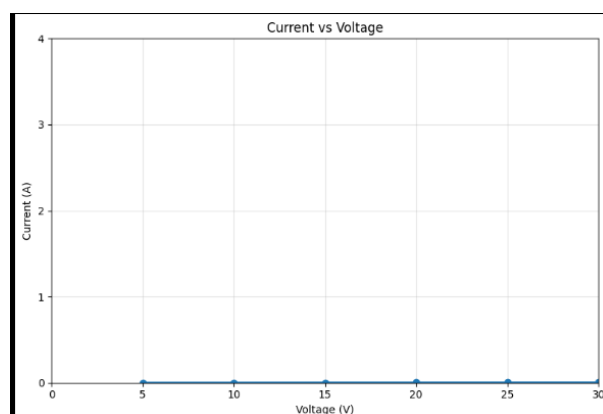
Since the measured current is approximately zero,

$$\boxed{I_D \approx 0\text{ mA}}$$

Similarly, for the other applied voltages, the diode current remains approximately zero.

3.5.2 Observation(includes lear graph)

The reverse-bias characteristics show that the reverse current of the PN junction diode is approximately zero for the applied voltage range used in the experiment. This indicates that the diode offers very high resistance under reverse bias.

**Figure 7**

Thus,

$$I_D \approx 0$$

and the diode behaves approximately as an open circuit in the reverse bias region before breakdown.

4 Part C - Zener Diode: Forward and Reverse Bias

4.1 Components Used

- Regulated Power Supply
- Zener Diode
- Resistor ($R = 1 \text{ k}\Omega$)
- Milliammeter
- Voltmeter

4.2 Forward bias

4.2.1 Circuit

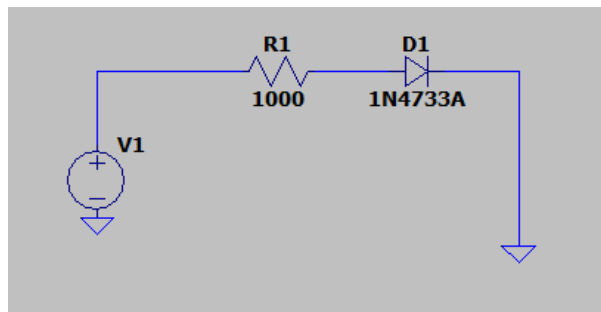


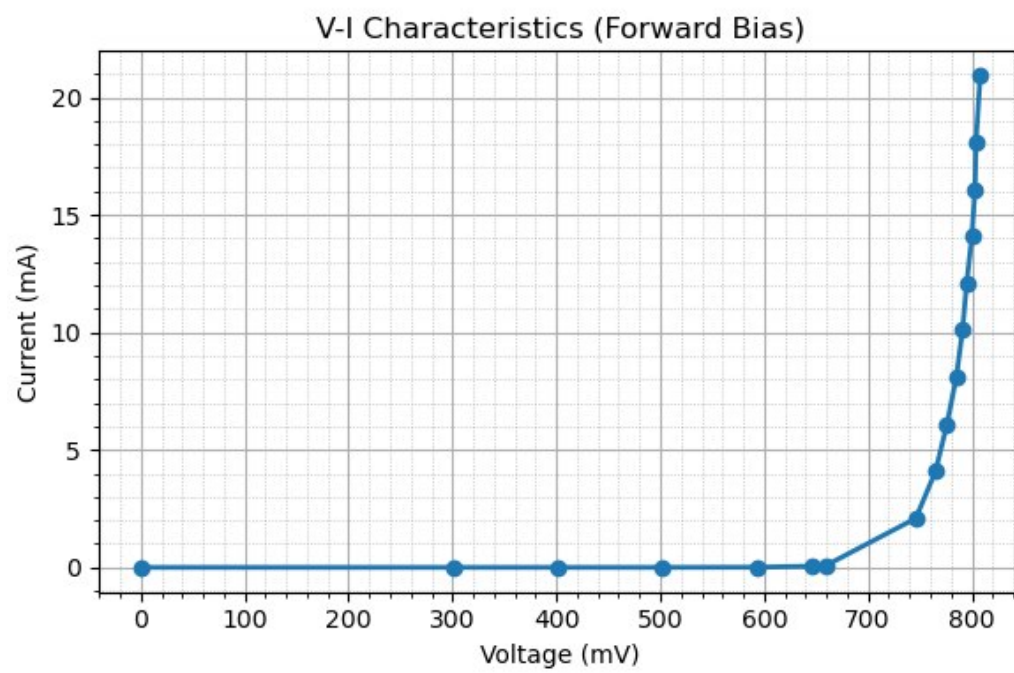
Figure 8: Zener diode - Forward bias

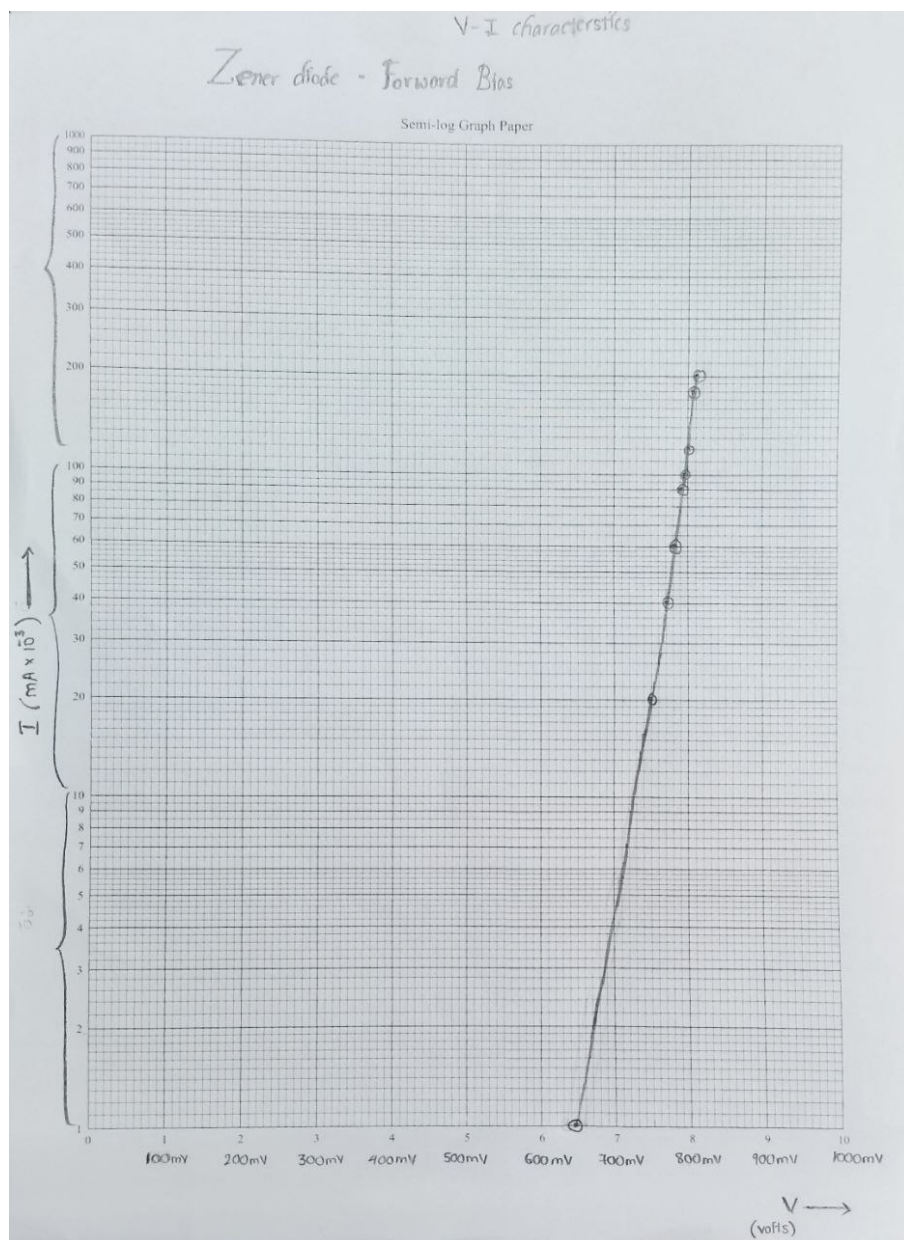
4.2.2 Observation Table

S.No.	Voltage (V)	Current (mA)
1	0.301	0
2	0.401	0
3	0.594	0.0085
4	0.660	0.0918
5	0.746	2.088
6	0.765	4.089
7	0.776	6.090
8	0.785	8.089
9	0.791	10.092
10	0.795	12.090
11	0.803	16.092
12	0.808	20.092

Table 3: Zener diode – Forward bias

4.2.3 Plots



**Figure 9:** $\log(I)$ vs V

4.2.4 Simulations

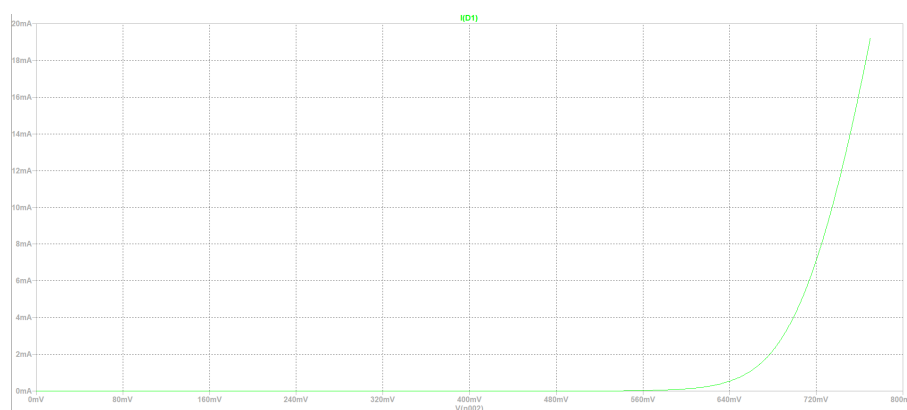


Figure 10: Simulation Plot - Forward bias (Zener diode)

The Zener diode characteristics were simulated in LTspice using a **1N4733A (5.1 V)** Zener diode model. A DC Sweep Analysis (.dc V1 0 20 0.01) was performed by varying the input voltage over a suitable range to obtain the forward V–I characteristics. The current through the diode and the voltage across it were observed and plotted.

4.2.5 Analysis

When forward-biased, a Zener diode acts just like a standard silicon PN-junction diode. Once the applied voltage reaches the forward knee threshold (around 0.7V for silicon), the current shoots up exponentially with even a tiny increase in voltage. So, even though Zener diodes are specifically for reverse breakdown, their forward behavior is completely ordinary

4.3 Reverse bias

4.3.1 Circuit

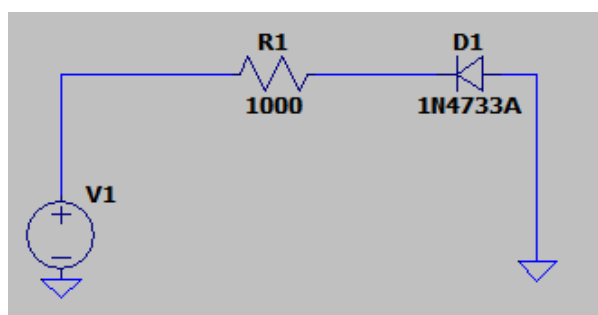


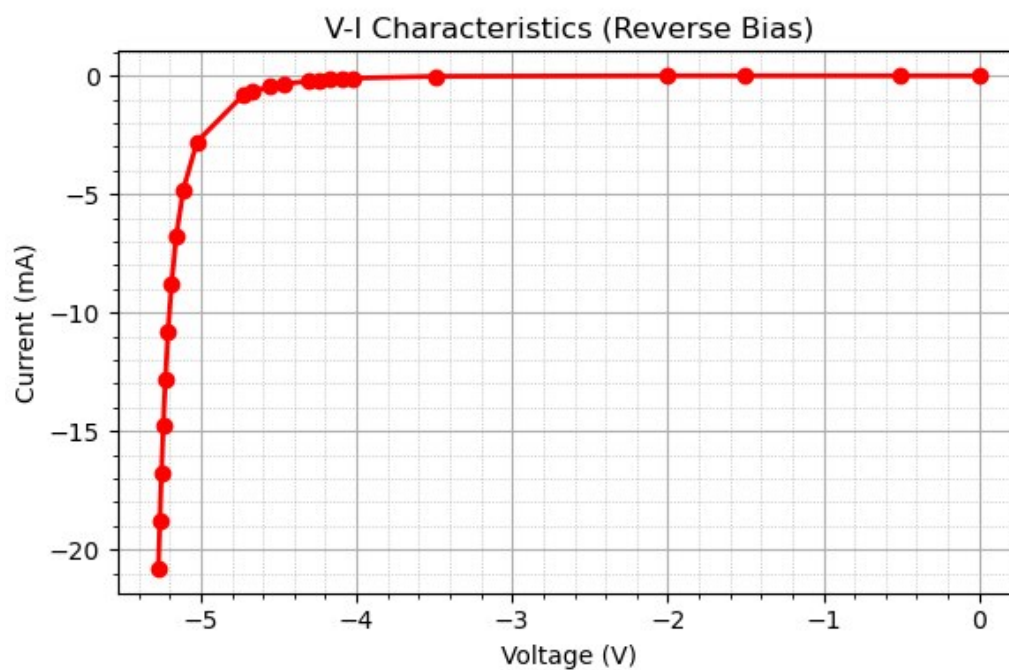
Figure 11: Zener diode - Reverse bias

4.3.2 Observation Table

S.No.	Voltage (V)	Current (mA)
1	-0.502	0
2	-2.010	0
3	-3.495	-0.023
4	-4.097	-0.101
5	-4.238	-0.184
6	-4.558	-0.464
7	-4.725	-0.797
8	-5.024	-2.789
9	-5.112	-4.799
10	-5.158	-6.802
11	-5.188	-8.800
12	-5.210	-10.800
13	-5.227	-12.790
14	-5.252	-16.795
15	-5.263	-18.800
16	-5.272	-20.795

Table 4: Zener diode – Reverse bias

4.3.3 Plots



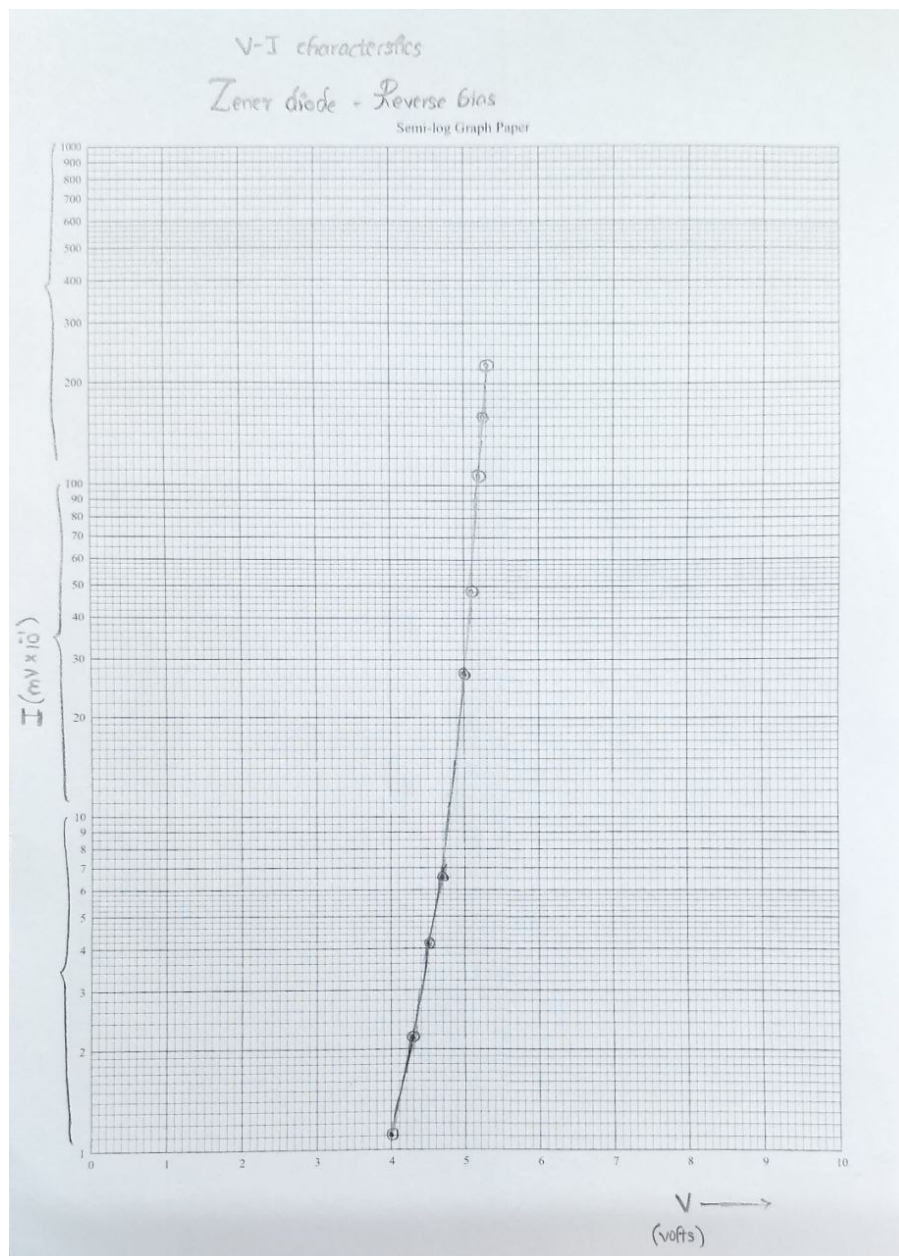


Figure 12: Enter Caption

4.3.4 Simulations

The Zener diode characteristics were simulated in LTspice using a **1N4733A** Zener diode connected in series with a $1\text{ k}\Omega$ resistor. Initially, a DC sweep from 0 V to 4 V with a step size of 0.5 V was performed to observe the reverse-bias region before breakdown.

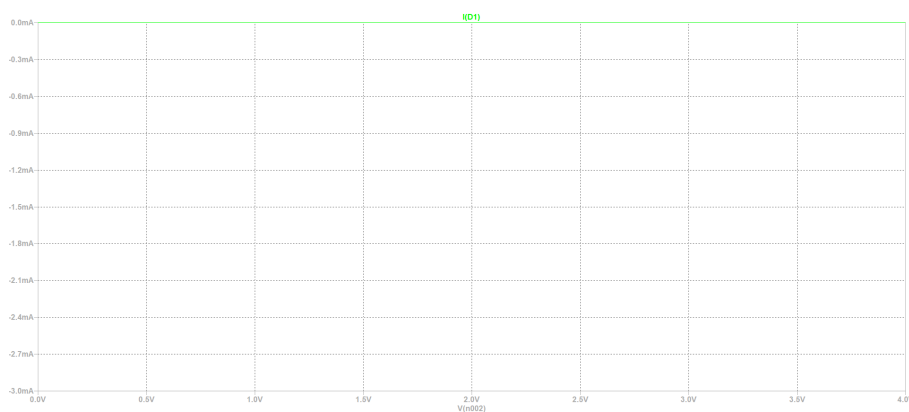


Figure 13: 0 to 4 with step size 0.5

To accurately capture the breakdown knee, a finer DC sweep from 4 V to 5.5 V with a step size of 0.1 V was carried out.

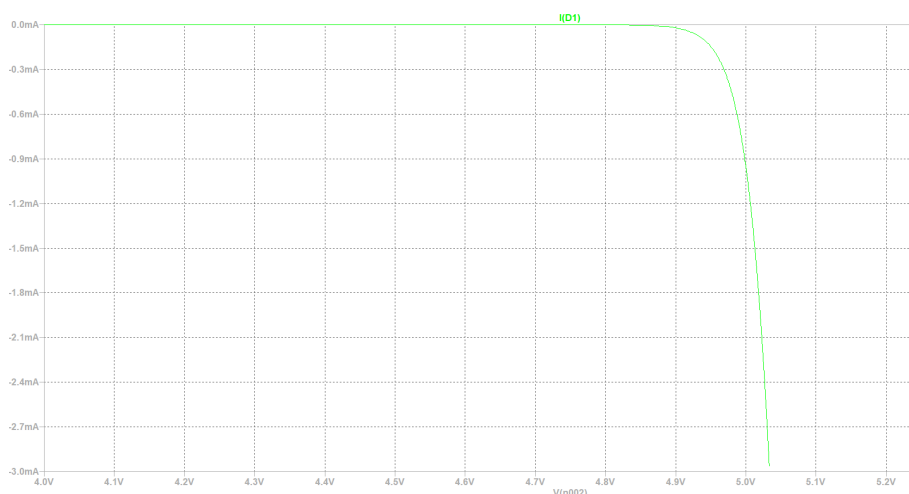


Figure 14: 4 to 5.5 with step size 0.1

Finally, a complete DC sweep from 0 V to 6 V with a step size of 100 mV was performed to obtain the overall reverse V–I characteristic of the Zener diode and clearly visualize the breakdown behavior.

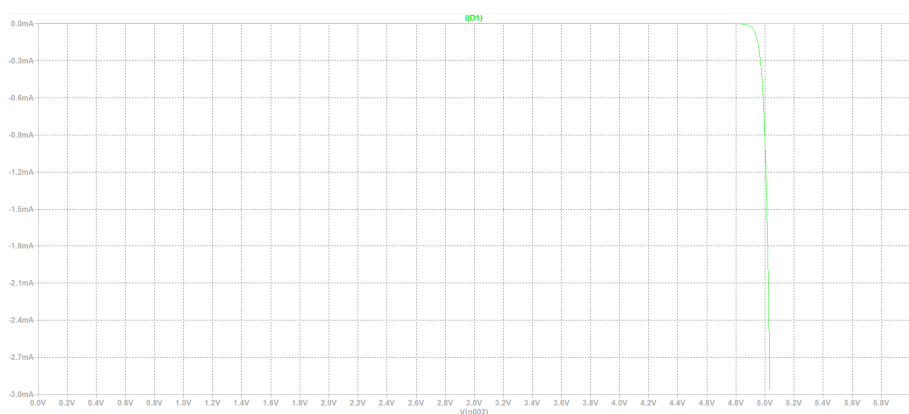


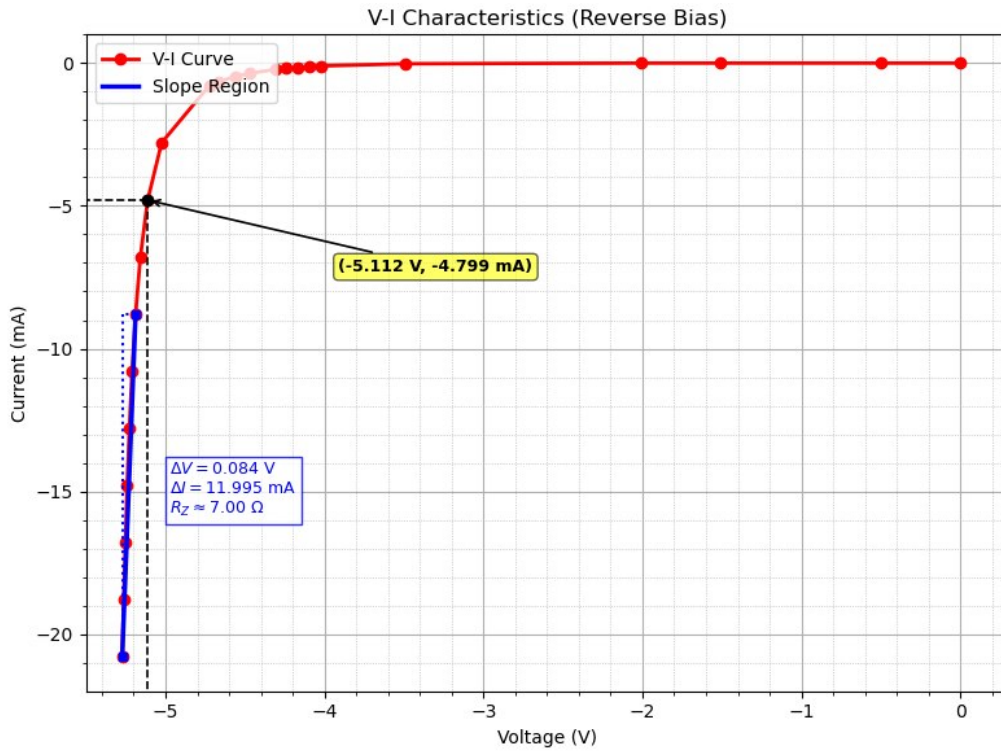
Figure 15: Simulation Plot - Reverse bias (Zener diode)

Theoretically, the reverse characteristic of a Zener diode lies in the third quadrant because the diode voltage V_r and reverse current are both negative. In this LTspice simulation, the voltage appears positive because the cathode node voltage is measured with respect to ground, while the current is negative since LTspice defines positive diode current from anode to cathode. During Zener breakdown, current flows from cathode to anode, resulting in a negative current.

4.3.5 Calculations and Analysis

From the V - I characteristics obtained from the experimental data, the reverse breakdown region is observed at approximately -5.112 V. Since the anode of the Zener diode is connected to ground ($V_G = 0$ V), the voltage across the diode in the reverse direction is -5.112 V. Hence, the Zener voltage, expressed as a positive magnitude, is

$$V_Z = 5.112 \text{ V.} \quad (1)$$



To determine the **dynamic Zener resistance**, two points well within the approximately linear breakdown region were selected from the data, $(-5.188 \text{ V}, -8.800 \text{ mA})$ and $(-5.272 \text{ V}, -20.795 \text{ mA})$. These points were chosen because they are sufficiently separated while remaining within the breakdown region, providing a more reliable estimate of the slope and minimizing the effect of rounding errors. Thus,

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{(-5.272) - (-5.188)}{(-20.795) - (-8.800)} \times 10^3 \approx 7.0 \Omega. \quad (2)$$

Therefore, the obtained values are

$$\boxed{V_Z = 5.112 \text{ V}}, \quad \boxed{r_Z \approx 7.0 \Omega}. \quad (3)$$

5 Part D - Clipper Circuits

5.1 Components Used

The following components were used to construct and simulate the clipper circuit:

- PN junction diode
- Resistor
- AC voltage source
- Connecting wires

5.2 Circuit

5.2.1 negative clipper

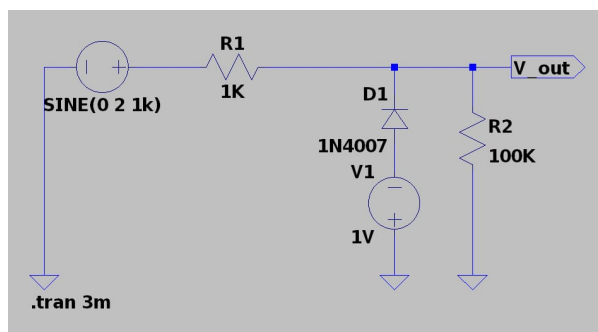


Figure 16

5.2.2 biased clipper

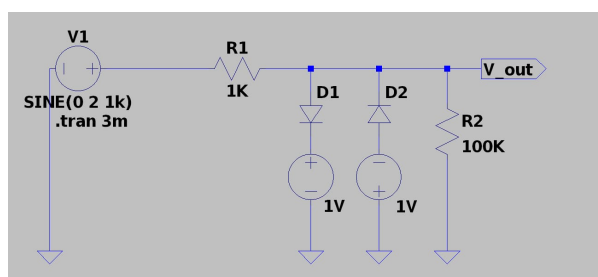


Figure 17

5.2.3 positive clipper

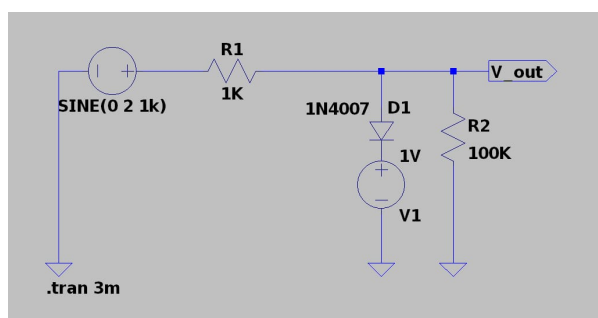


Figure 18

5.3 Observations

The input and output voltages of the clipper circuit were measured for different values of the input voltage.

5.3.1 negative clipper

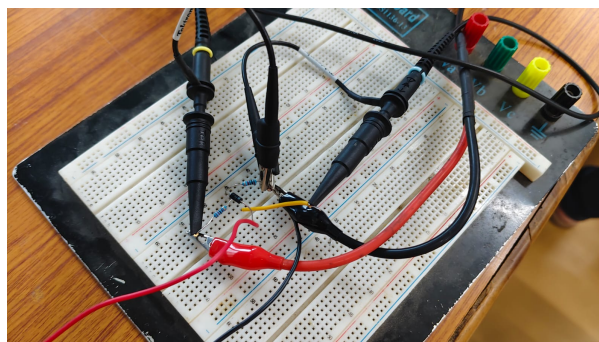


Figure 19

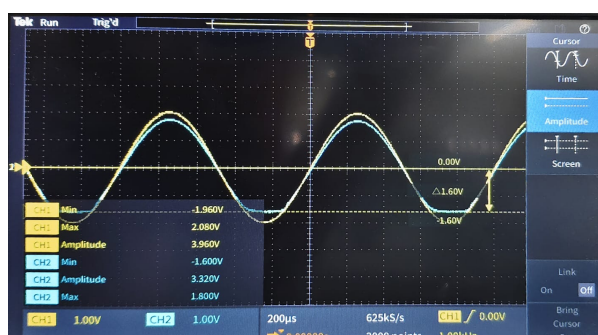


Figure 20

5.3.2 biased clipper

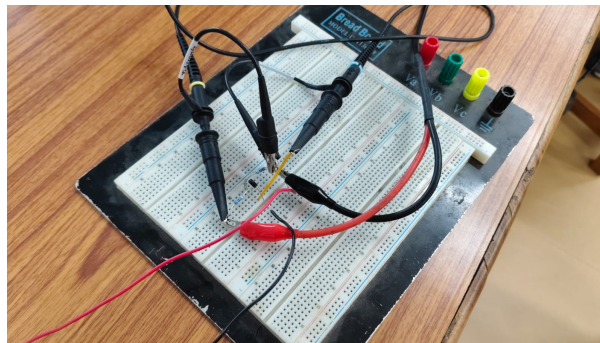


Figure 21

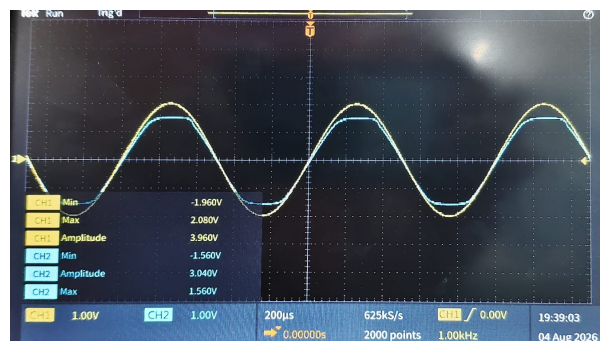


Figure 22

5.3.3 positive clipper

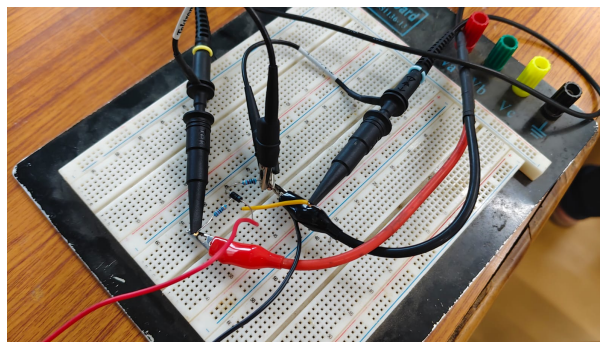


Figure 23

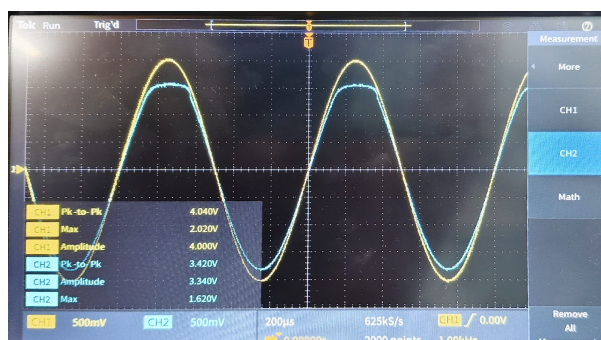


Figure 24

5.4 Simulations

5.4.1 negative clipper

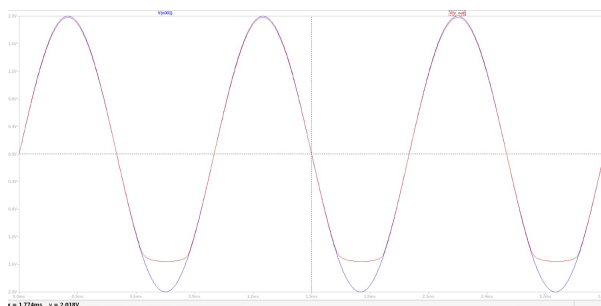


Figure 25

5.4.2 biased clipper

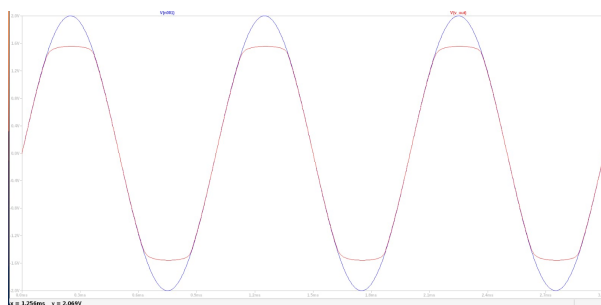


Figure 26

5.4.3 positive clipper

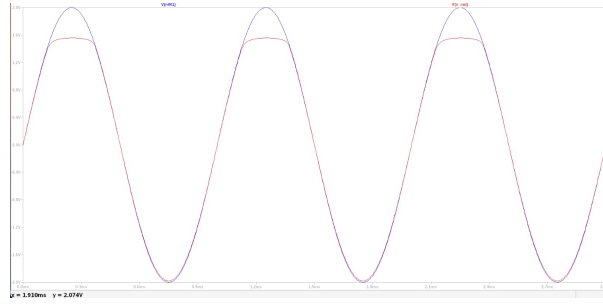


Figure 27

5.5 Calculations and Analysis

For the silicon PN junction diode used in the clipper circuit, the forward voltage drop is approximately

$$V_D \approx 0.6 \text{ V}.$$

Therefore, the clipping voltage is

$$V_{\text{clip}} \approx 0.6 \text{ V}.$$

5.5.1 Case 1: Diode ON

When the input voltage is sufficiently high to forward bias the diode,

$$V_{\text{in}} > V_D,$$

the diode conducts. For the diode used in the experiment,

$$V_D \approx 0.6 \text{ V}.$$

Therefore, the output voltage is limited approximately to the clipping level:

$$V_{\text{out}} \approx 0.6 \text{ V}.$$

The current through the series resistor can be calculated using Ohm's law:

$$I = \frac{V_{\text{in}} - V_D}{R}.$$

For example, if

$$V_{\text{in}} = 2 \text{ V}$$

and

$$R = 1 \text{ k}\Omega,$$

then

$$I = \frac{2 - 0.6}{1000}$$

$$I = 1.4 \times 10^{-3} \text{ A}$$

$$\boxed{I = 1.4 \text{ mA}}.$$

5.5.2 Case 2: Diode OFF

When the input voltage is below the forward-bias voltage,

$$V_{\text{in}} < 0.6 \text{ V},$$

the diode remains OFF. Therefore,

$$I \approx 0.$$

Since the current through the series resistor is approximately zero,

$$V_R = IR \approx 0.$$

Hence, the output voltage approximately follows the input voltage:

$$\boxed{V_{\text{out}} \approx V_{\text{in}}}.$$

Thus, the input waveform is passed to the output without significant clipping in this region.

5.5.3 Analysis

Therefore, the clipper circuit operates in two regions:

$$\boxed{\begin{cases} V_{\text{out}} \approx V_{\text{in}}, & \text{Diode OFF, } V_{\text{in}} < 0.6 \text{ V} \\ V_{\text{out}} \approx 0.6 \text{ V}, & \text{Diode ON, } V_{\text{in}} > 0.6 \text{ V} \end{cases}}$$

Hence, the portion of the input waveform above approximately 0.6 V is clipped.

The LTspice simulation shows that when the input voltage exceeds the forward voltage of the diode, the diode starts conducting and the output voltage remains approximately at the clipping level. Any additional increase in input voltage mainly appears across the series resistor, resulting in an increase in diode current.

The slight difference between the theoretical clipping voltage of 0.6 V and the simulated value can be attributed to the non-ideal voltage-current characteristic of the practical PN junction diode.

6 Part E - Clamper Circuits

A clamper circuit shifts an AC waveform vertically by adding a DC level without changing its shape. It uses a diode and capacitor, where the capacitor charges to nearly the peak input voltage when the diode conducts. The stored charge then shifts the waveform during the rest of the cycle. A positive clamper shifts the waveform upward, while a negative clamper shifts it downward.

6.1 Components Used

- PN junction diode
- Resistor ($R = 100 \text{ k}\Omega$)
- Capacitor ($C = 0.1 \text{ }\mu\text{F}$)
- Function generator
- DSO
- Connecting wires

6.2 Positive Clamper

6.2.1 Circuit

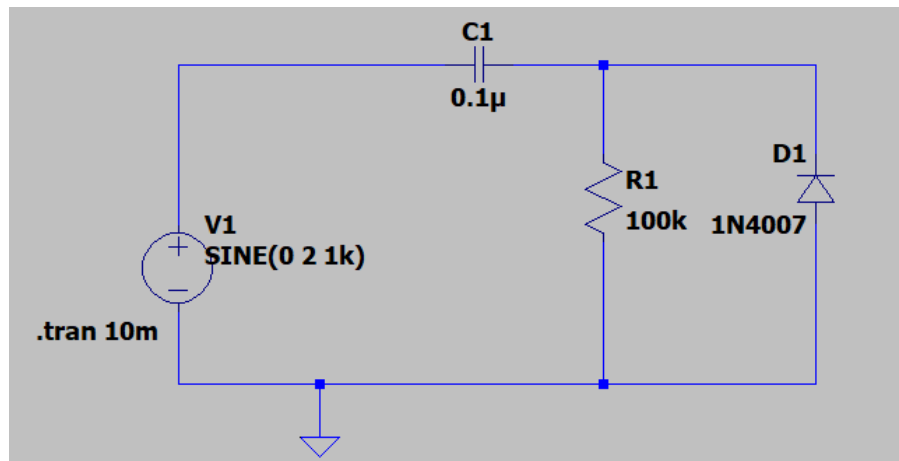
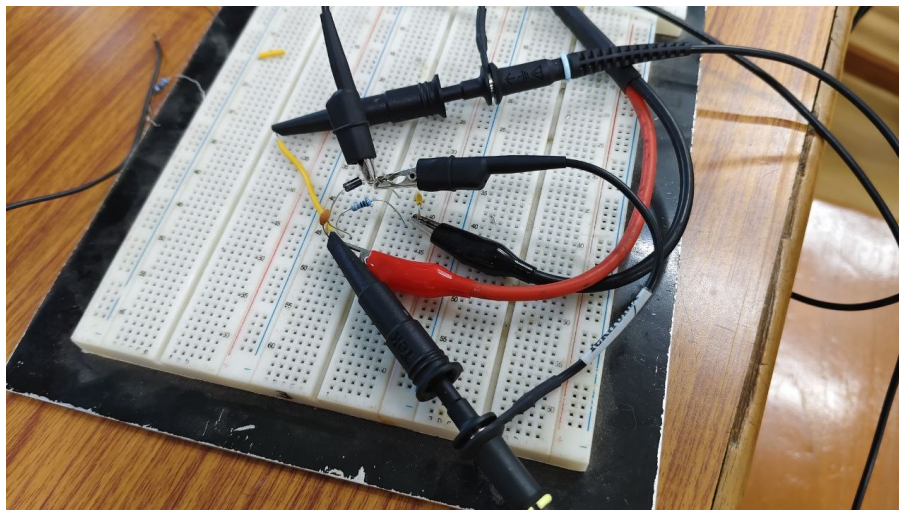
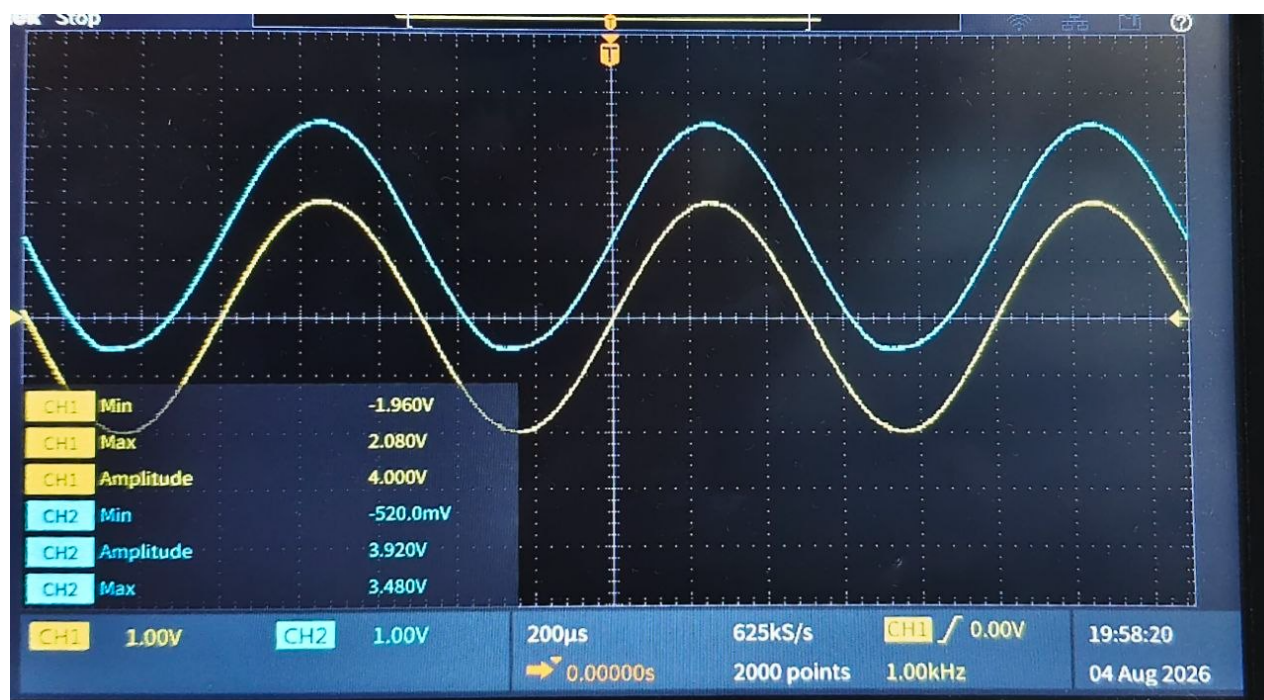


Figure 28: Positive clamper circuit

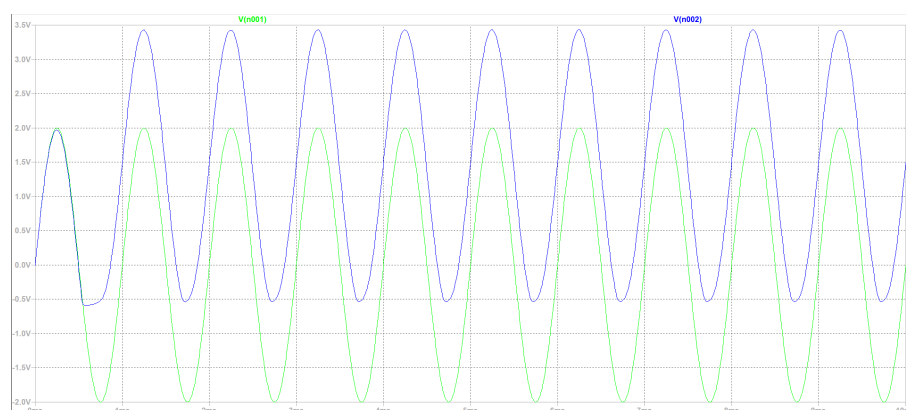


6.2.2 Observations



6.2.3 Simulations

The transient simulation of the positive clamper circuit is performed using a sine-wave input of 2 V peak amplitude at a frequency of 1 kHz, corresponding to 4 V_{pp}. The transient analysis is carried out for 10 ms, covering approximately 10 complete cycles. In the plot, the **green waveform** represents the input voltage V_{in} , while the **blue waveform** represents the output voltage V_{out} . The output waveform retains the same frequency and shape as the input but is shifted upward due to the clamping action.



6.2.4 Calculations and Analysis

The input signal is a sine wave of frequency 1 kHz with a peak-to-peak voltage of 4 V and zero DC offset.

$$V_{\text{in(pp)}} = 4 \text{ V} \quad (4)$$

Therefore, the peak value of the input voltage is

$$V_p = \frac{V_{\text{in(pp)}}}{2} = \frac{4}{2} = 2 \text{ V}. \quad (5)$$

For a positive clamper, the theoretical DC shift is approximately equal to the peak input voltage minus the forward voltage drop of the diode:

$$V_{\text{shift,expected}} = V_p - V_D. \quad (6)$$

Taking the forward voltage drop of the silicon diode as $V_D \approx 0.7 \text{ V}$,

$$V_{\text{shift,expected}} = 2 - 0.7 = 1.3 \text{ V}. \quad (7)$$

From the DSO observation,

$$V_{\text{in,min}} = -1.960 \text{ V} \quad (8)$$

and

$$V_{\text{out,min}} = -0.520 \text{ V}. \quad (9)$$

For a positive clamper, the measured DC shift is obtained from the difference between the minimum input and output voltage levels:

$$V_{\text{shift,measured}} = V_{\text{out,min}} - V_{\text{in,min}}. \quad (10)$$

Hence,

$$V_{\text{shift,measured}} = (-0.520) - (-1.960) = 1.440 \text{ V}. \quad (11)$$

The absolute error is calculated as

$$\text{error} = |V_{\text{shift,measured}} - V_{\text{shift,expected}}| \quad (12)$$

Therefore,

$$\text{error} = |1.440 - 1.300| = 0.140 \quad (13)$$

Circuit	Vin (Vpp)	Expected DC shift	Measured DC shift	Error
Positive	4.00	1.300	1.440	0.140

- The measured DC shift for the positive clamper was 1.44 V, whereas the theoretical value was approximately 1.30 V, resulting in a slight deviation.
- The deviation can be attributed to the non-ideal forward voltage of the silicon diode, which may not be exactly 0.7 V under the actual operating conditions.
- The capacitor may not charge to exactly the ideal peak input voltage due to the practical charging process and capacitor leakage.
- Component tolerances and uncertainty in determining the exact minimum voltage levels from the DSO also contribute to the deviation.

6.3 Negative Clamper

6.3.1 Circuit

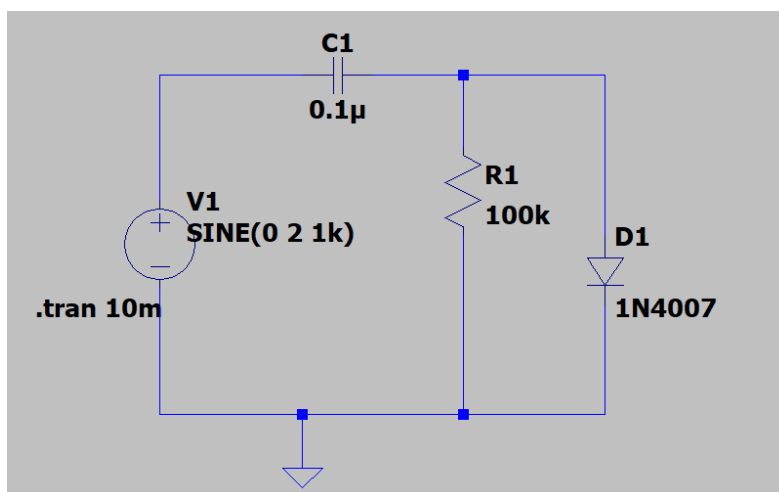
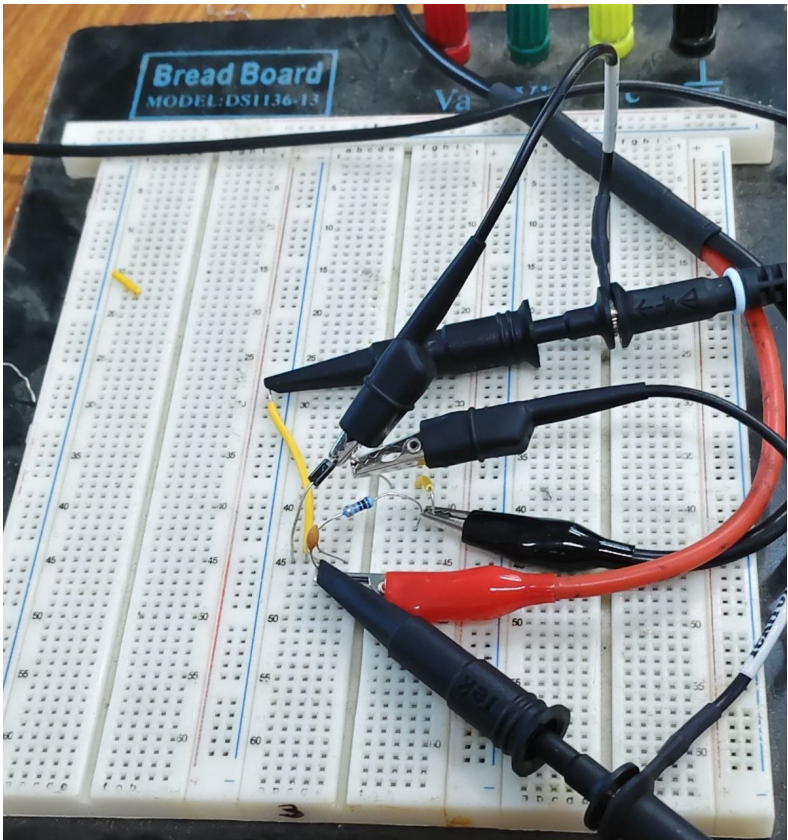
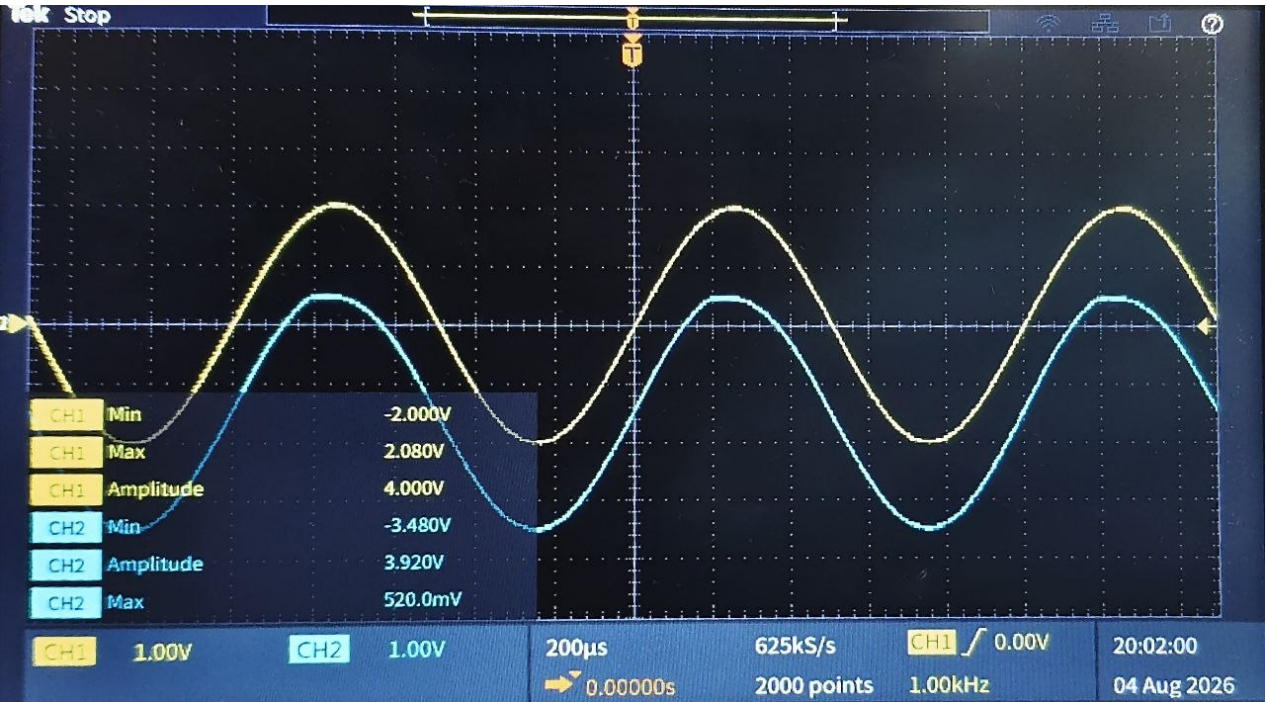


Figure 29: Negative clamper circuit

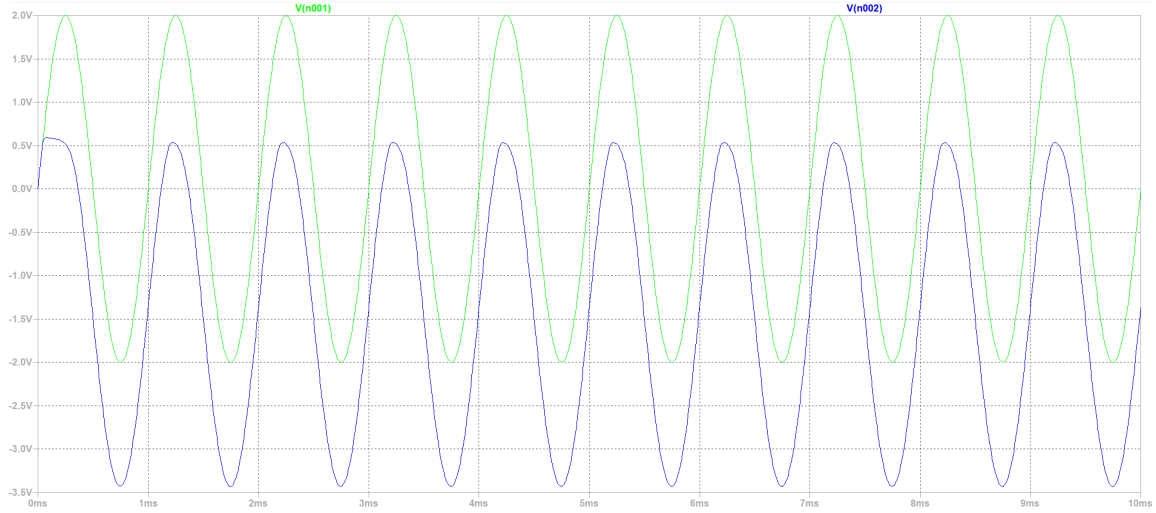


6.3.2 Observation Table



6.3.3 Simulations

The transient simulation of the clamper circuit uses a sine-wave input of 2 V peak amplitude at 1 kHz, corresponding to 4 V_{pp}. The transient analysis is performed for 10 ms, covering approximately 10 complete cycles. In the plot, the **green waveform** represents the input voltage V_{in} , while the **blue waveform** represents the output voltage V_{out} . The output retains the same frequency and shape as the input but is shifted downward.



6.3.4 Calculations and Analysis

The input signal is a sine wave of frequency 1 kHz with a peak-to-peak voltage of 4 V and zero DC offset.

$$V_{in(pp)} = 4 \text{ V} \quad (14)$$

Therefore, the peak value of the input voltage is

$$V_p = \frac{V_{in(pp)}}{2} = \frac{4}{2} = 2 \text{ V}. \quad (15)$$

For a negative clamper, the theoretical DC shift is approximately equal to the peak input voltage minus the forward voltage drop of the diode:

$$V_{\text{shift,expected}} = V_p - V_D. \quad (16)$$

Taking the forward voltage drop of the silicon diode as $V_D \approx 0.7 \text{ V}$,

$$V_{\text{shift,expected}} = 2 - 0.7 = 1.3 \text{ V}. \quad (17)$$

From the DSO observation,

$$V_{in,max} = 2.080 \text{ V} \quad (18)$$

and

$$V_{\text{out,max}} = 0.520 \text{ V}. \quad (19)$$

For a negative clamper, the measured DC shift is obtained from the difference between the maximum input and output voltage levels:

$$V_{\text{shift,measured}} = V_{\text{in,max}} - V_{\text{out,max}}. \quad (20)$$

Hence,

$$V_{\text{shift,measured}} = 2.080 - 0.520 = 1.560 \text{ V}. \quad (21)$$

The absolute error is calculated as

$$\text{error} = |V_{\text{shift,measured}} - V_{\text{shift,expected}}| \quad (22)$$

Therefore,

$$\text{error} = |1.560 - 1.300| = 0.26 \quad (23)$$

Circuit	V _{in} (V _{pp})	Expected DC shift	Measured DC shift	Error
Negative	4.00	1.300	1.560	0.260

- The measured DC shift for the negative clamper was 1.56 V, compared with the theoretical value of 1.30 V, resulting in a larger deviation than that observed for the positive clamper.
- The deviation can be attributed to practical non-idealities such as the actual forward voltage of the diode, capacitor leakage, component tolerances, and uncertainty in measuring the maximum voltage levels using the DSO.
- Therefore, the difference between the measured and theoretical values is more likely due to the non-ideal diode forward voltage, capacitor and component characteristics, and measurement uncertainty.
- The slight asymmetry in the input waveform, with $V_{\text{in,max}} = 2.08 \text{ V}$ and $V_{\text{in,min}} = -2.00 \text{ V}$, may also contribute to the difference in the measured shift.

7 Conclusion

- The V–I characteristics of the PN junction diode and Zener diode were studied experimentally and through LTspice simulations under forward and reverse bias conditions.
- The clipping and clamping circuits were also studied, and their output waveforms were observed to verify the respective amplitude limiting and DC level-shifting operations.